

Jason Dsouza

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PROFESSIONAL SUMMARY

Geologist and GIS Specialist with 2 years building spatial data pipelines, deploying cloud-based GIS infrastructure, and delivering analysis-ready geospatial data for AI-powered projects. Skilled in QGIS, Python (GeoPandas, Rasterio, GDAL, etc.), PostgreSQL, PostGIS, REST APIs, and cloud-based GIS infrastructure (GeoServer on AWS). Experienced in web scraping, WMS/WFS service deployment on GeoServer, and agentic AI workflows using tools such as Claude Code, Gemini CLI, and Antigravity. Passionate about engineering scalable, automated geospatial solutions and translating complex domain data into queryable, structured formats.

TECHNICAL SKILLS

- **GIS Software:** QGIS (Advanced), ArcGIS, GeoServer, Google Earth Engine (Python API), GDAL
- **Programming:** Python (Advanced), SQL, Bash/Shell Scripting
- **Geospatial Libraries:** GeoPandas, Shapely, Rasterio, GDAL, Fiona, OWSLib
- **Databases:** PostgreSQL, PostGIS, supabase, MongoDB
- **Data Acquisition:** Selenium, Requests, BeautifulSoup, REST APIs
- **Cloud & DevOps:** AWS EC2, S3, Linux, Git/GitHub, Docker
- **AI and Agents:** Agentic AI development using Claude Code, Gemini-CLI, Antigravity, Gemini API
- **Domain Knowledge:** Remote Sensing, Photogrammetry, Geological Mapping, Spatial Analysis

WORK EXPERIENCE

RadiXplore | *Digital Earth Scientist (GIS) & Business Development Lead* | May 2023 - Jan 2025

- Built Python-based GIS data pipelines to scrape, clean, and standardize large spatial datasets from government geological portals, producing analysis-ready layers compatible with database workflows and downstream AI processing.
- Deployed and administered GeoServer spatial services (WMS/WFS) on AWS EC2, backed by PostGIS for geometry storage and spatial queries, delivering stable geographic data services to internal teams and product applications.
- Processed and quality-controlled spatial datasets using GeoPandas and Shapely, correcting geometric errors, resolving projection inconsistencies, and enforcing schema standards across large datasets.
- Converted thousands of unstructured geological documents into structured geospatial layers by linking textual location references to spatial coordinates, making domain knowledge queryable.
- Automated data acquisition workflows using Python web scraping libraries (Selenium, Requests, BeautifulSoup), eliminating manual download processes across multiple web portals.
- Automated web-app testing using workflows like Playwright and Selenium.
- Standardized Jira project management by designing a best-practice schema for ticket creation and workflows across multiple cross-functional teams.
- Bridged technical gaps by writing clear Confluence documentation, enabling non-GIS personnel to independently configure and operate Web-GIS pipelines.

PROJECTS

GeoBox-AR | *Python, OpenCV, Computer Vision, Hardware Integration, Linux*

- Designed and built a low-cost Augmented Reality Sandbox for the Geology Department at St. Xavier's College, currently in active use for student teaching.
- Developed the core software in Python using OpenCV (computer vision) to translate physical terrain interactions into real-time digital overlays, making geological concepts interactive without expensive commercial hardware.
- Used AI Agents like Claude Code for quick prototyping and frontend design
- Managed the full project scope including hardware sourcing, software development, faculty coordination, and deployment.
- Utilized open-source libraries including PySide6/PyQt (GUI), OpenCV, NumPy, and Libfreenect for data processing, making sure the software product remains cross-platform for Windows, Linux and MacOS users.
- [Video Demo](#)
- [LinkedIn Article](#)
- [Blog Article](#)
- [Website \(in progress\)](#)

Automatic Georeferencer | *Python, OpenCV, Multi-thread Computing*

- Built a completely offline automatic georeferencer for SOI 50K Toposheets, successfully georeferencing 250+ toposheets in under 7 minutes using parallel processing.
- Engineered an automated workflow using Pathlib and regex to index files, GeoPandas to load spatial boundaries, OpenCV to crop images to map frames, and Rasterio to assign coordinates to map corners. Planning to launch this as a QGIS plugin, soon.
- [LinkedIn Article](#)

A.I. Petrographer | *Python, Streamlit, Gemini-API*

- Built a web-app in Python where the user can upload Plane-Polarised and Cross-Polarised images of rock thin-sections to get A.I. insights about mineralogy, texture, rock name and more.
- Used Gemini API with a petrology-oriented system prompt for the LLM, Streamlit for GUI, file management and webhosting.
- [Project Link](#)

GIS Tools (Web-App) | *Python, Streamlit*

- Built a web-app in Python for on-the-go GIS visualisation and analysis.
- Current features include:-
 - Shapefile viewer that allows the user to drag and drop shapefiles, GeoJSONs and other spatial vector files onto a map and view them without having to load up desktop softwares.
 - Plot WMS layers
 - Download OSM features of an area by drawing a bounding box on a map
- Used Streamlit for webhosting, Folium for the map components, GeoPandas to handle vector data, OWSLib for fetching WMS layers.
- [Project Link](#)

The GIS Toolkit | *Blog Writing*

- Authored practical tutorials on open-source GIS workflows and spatial data engineering on a personal blog.
- Created content showing how geoscientists can transition from manual GIS tools to programmatic pipelines and leverage new advances in GIS and Geotech.
- [Project Link](#)

EDUCATION

Master of Science (M.Sc.) in Geology | St. Xavier's College, Mumbai | June 2024 - April 2026

- **Dissertation:** GeoBox-AR - Augmented reality framework for geoscience education.
- **Coursework:** GIS, remote sensing, Stratigraphy, Petrology, Economic Geology, Engineering Geology, Hydrogeology, Oceanography, Petroleum and Mining Geology

Bachelor of Science (B.Sc.) in Geology | St. Xavier's College, Mumbai | Aug 2020 - June 2023

CERTIFICATIONS

- ESRI Cartography Course (Esri)
- AI Aware & AI Appreciate Badges (AI For All)